

FeelAI

Emotion Mapping Through HR and EDR

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Which sensors did we use?

Our system operates through a comprehensive five-stage pipeline, starting with data collection using PulseSensor for heart rate (HR) and Grove GSR Sensor for electrodermal resistance (EDR). These signals are captured via an Arduino Nano board and transmitted through serial communication.

Why these variables?

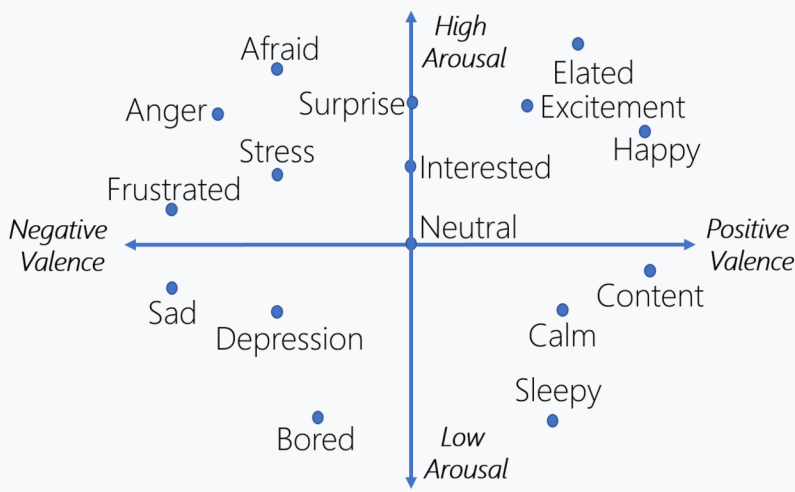
Heart rate reflects autonomic arousal and electrodermal response captures sympathetic skin conductance, giving complementary physiological markers closely linked to emotional valence and intensity. Both can be sensed cheaply and non-invasively with off-the-shelf modules on Arduino or by hardware employed by many popular models of smartwatches and smart rings, enabling reliable real-time emotion prediction.

What data do we use?

To train the model we used the publicly available WESAD multimodal physiological dataset from the University of Siegen. Its wrist based HR and EDR signals with aligned emotion labels match our sensing setup, giving a realistic training and evaluation benchmark.

Our data pipeline

We cut the HR and EDR signals into 15 second windows, extracted HR features from each HR slice with HeartPy and computed mean, standard deviation, maximum and minimum of the corresponding EDR segment. The most frequent label inside each window becomes its target class, and windows from all subjects are merged into one table after discarding the rarely used labels 5–7. Rows containing NaNs are dropped, then we create stratified splits: 80 % train, 10 % validation and 10 % test, preserving the emotion distribution. Before modelling we cast every feature to float, remove auxiliary columns such as subject IDs and specific HRV ratios, leaving a clean numeric matrix ready for optional SMOTE balancing and XGBoost training.



Introduction

FeelAI is an innovative emotion-tracking application designed to leverage existing wearable devices, such as smartwatches and rings, to monitor emotions through biometric signals. By analyzing heart rate (HR) and electrodermal resistance (EDR), our solution classifies emotional states based on valence and arousal. It's based on a concept of emotion matrix, represented in the graph above. Through advanced signal filtering, precise feature extraction, and sophisticated machine learning techniques, FeelAI offers users real-time emotional feedback and provides actionable alerts to mental health professionals, enhancing overall well-being without the need for specialized hardware.

TRAINING AND MODELING

The core of our training methodology involved an iterative hyperparameter tuning process to identify the optimal configuration for our machine learning models. For each algorithm, we defined a grid of key parameters, such as max_depth, learning_rate, subsample, and regularization terms. We then trained multiple versions of the model on the training dataset, each with a different combination of these parameters. The performance of each version was subsequently measured on the validation set, using metrics like F1-score and AUC to assess its predictive accuracy. The set of hyperparameters that yielded the highest performance on the validation data was selected for the final model, which was then tested on the holdout test set to confirm its effectiveness.

Throughout our experimentation, we evaluated several advanced machine learning algorithms, beginning with Gradient Boosting implementations like XGBoost. After a thorough comparison of results, CatBoost was ultimately selected as our final model, as it demonstrated superior performance in classifying emotional states on our validation dataset.

Metric	Value
Accuracy	0.6424
Error Rate	0.3576
F1-Score (Macro)	0.6113
Precision (Macro)	0.6419
Recall (Macro)	0.5923
AUC (Macro)	0.8636

Table 1. Summary of the error metrics of the model on the test data.

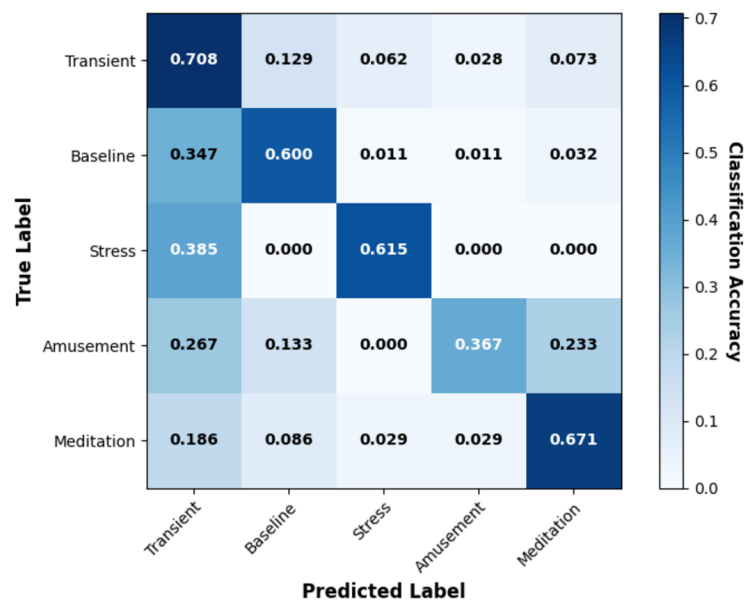


Figure 1. Confusion matrix. The heatmap compares true and predicted classes for the five target states. Values close to 1 along the diagonal indicate solid recognition of states Transient and Meditation, whereas states Amusement and Baseline exhibit spill-over into neighbouring categories, exposing where physiological signatures overlap.

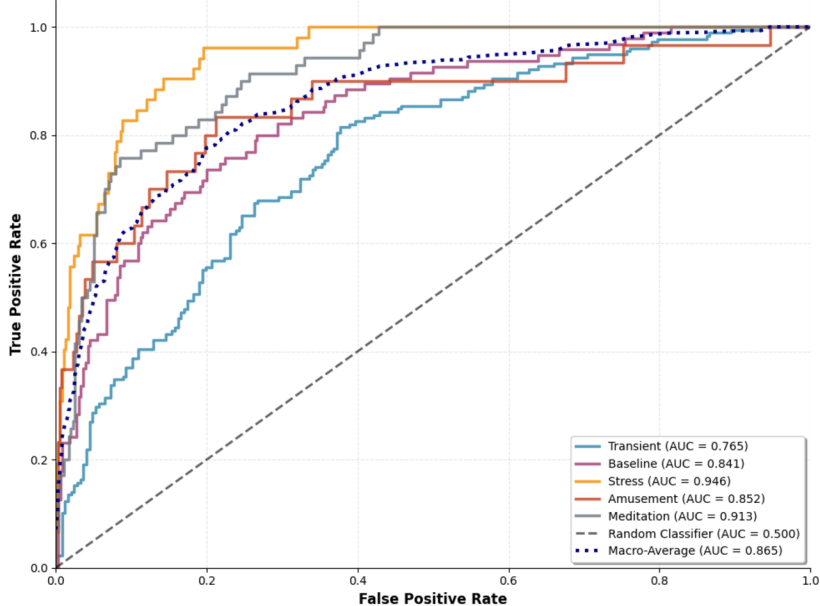


Figure 2. ROC curves. The centre plot traces one-versus-rest performance across thresholds. All classes achieve high separability with AUCs from roughly 0.77 to 0.95 and a macro-average near 0.87, showing that the classifier maintains strong discrimination even when operating at different sensitivity levels.

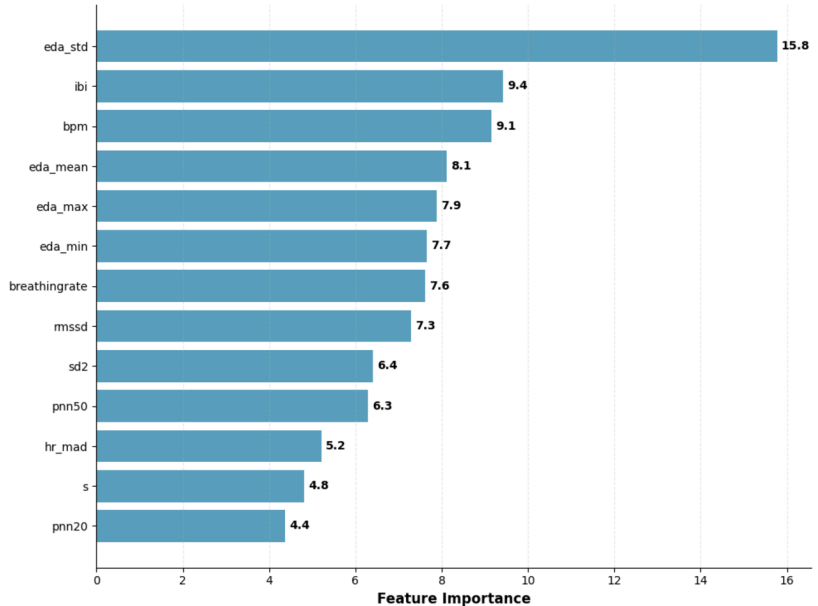


Figure 3. Feature importance. The bar chart ranks predictors by their contribution to XGBoost gain. Electrodermal variability (eda_std) dominates, followed by inter-beat interval and mean EDA metrics, implying that sympathetic arousal and heart-rate timing jointly underpin the model's decisions – consistent with the patterns highlighted in the confusion matrix and ROC analysis.

Results

Our experiments confirm that off-the-shelf wearables can double as emotion sensors. Streaming 64 Hz PPG beats and 4 Hz skin resistance readings through a five-stage pipeline, artifact removal, 15 second windowing, HeartPy-based HR feature engineering, CatBoost ensemble tuning, and post-hoc evaluation that let us separate five affective conditions in the WESAD benchmark with 64 % accuracy, 0.61 macro-F1, and 0.86 macro-AUC.

Feature attribution pinpoints EDA variance (eda_std) and inter-beat dynamics (IBI, BPM) as the strongest physiological markers, proving that inexpensive sensors capture the autonomic signatures of stress, relaxation, and positive arousal. Because all computations finish well inside the 15-second window, the model is light enough for on-device inference, enabling real-time mood logging, proactive stress alerts, and one-tap escalation to mental-health professionals with keeping raw data on the user's wrist.

These results turn every smartwatch into a personal emotion co-processor. Upcoming steps would be to personalise the model with the transfer-learning to individual baselines, work on user data privacy via on-device training and differential-privacy noise and expand the beyond the valence-arousal plane to richer, discrete emotions and validated in clinical trials. In short, FeelAI bridges affordable consumer hardware and rigorous biosignal analytics to put mental-wellbeing insights literally at hand.



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